- f.c. repartiție (!) ← caz discret, caz continuu (user)
- f.c. / f. caracteristică
- f. generatoare / f. caracteristică

Azi: • [LNM - legi ale numerelor mari; TLC - teorema limită centrală] ← instrumente f. importante la stat. final
• vectori aleatori - vectori bidimensionali { covariația, coeficient de corelație
• exemple / ex. / proiect / antet formule

1) LNM

• Inegalitatea Cebâșev

X v.a. reală pt care $M(X)$, $D^2(X)$ finite (σ^2), $\sigma = \sqrt{D^2(X)}$ = abaterea medie pătratică

$$P(|X - M(X)| \leq \varepsilon) \geq 1 - \frac{D^2(X)}{\varepsilon^2} \quad (\varepsilon > 0).$$

$(m = M(X))$

caz special: $\varepsilon = 3\sigma \Rightarrow$

$$P(|X - m| \leq \varepsilon) \geq 1 - \frac{\sigma^2}{9\sigma^2} = 1 - \frac{1}{9} = \frac{8}{9}.$$

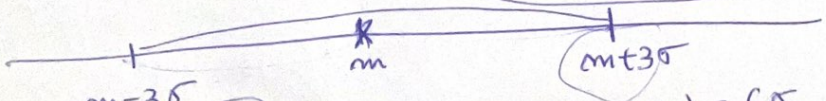
$$P(|X - m| \leq \varepsilon) \geq \frac{8}{9} \quad (\approx 0.9)$$

($\varepsilon = 3\sigma$)

$|x| \leq 3 \quad -3 \leq x \leq 3$

$|X - m| \leq \varepsilon = 3\sigma \quad -\varepsilon \leq X - m \leq \varepsilon$

$m - 3\sigma \leq X \leq m + 3\sigma$



$m - 3\sigma$

Regula celor 3σ //

$m + 3\sigma - (m - 3\sigma) = 6\sigma$

regula celor 6σ

Tipuri de convergență

★ convergență în probabilitate.

$(X_n)_n$ v.a., X v.a.

$X_n \xrightarrow{(P)} X$ dacă $\lim_{n\to\infty} P(|X_n - X| < \varepsilon) = 1$, $\forall \varepsilon > 0$.

- $X_n \xrightarrow{(P)} X$, $Y_n \xrightarrow{(P)} Y$ $\Rightarrow a X_n + b Y_n \xrightarrow{(P)} aX + bY$ $(a, b \in \mathbb{R})$
- $X_n \xrightarrow{(P)} X \Rightarrow |X_n| \xrightarrow{(P)} |X|$

★ convergență în repartiție

$(X_n)_n$, X v.a. F_n fc. repartiție pt X_n. F - - - - X.

$X_n \xrightarrow{rep.} X$ dacă F_n converge la F în orice punct de continuitate pt F.

- $X_n \xrightarrow{(P)} X \Rightarrow X_n \xrightarrow{rep.} X$
- $c \in \mathbb{R}$, $X_n \xrightarrow{rep.} c \Rightarrow X_n \xrightarrow{(P)} c$

★ convergență aproape sigură

$(X_n)_n$, X v.a., $X_n \xrightarrow{a.s.} X$ dacă $P\left(\omega \mid \lim_{n\to\infty} X_n(\omega) = X(\omega)\right) = 1$.

- $X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{(P)} X \Rightarrow X_n \xrightarrow{rep.} X$.
- $X_n \xrightarrow{(P)} X \Leftrightarrow (\forall)(X_{n_k}) \subset (X_n) \Rightarrow (\exists)(X_{n_{k'}}) \subset (X_{n_k})$ cu prop. $X_{n_{k'}} \xrightarrow{a.s.} X$

* Convergența în medie

$(X_n)_n$, X v.a. ;

$X_n \xrightarrow[\text{ordin } r]{\text{medie}} X$ dacă $\lim_{n\to\infty} M(|X_n - X|^r) = 0$.

$r=2$ ‖ $X_n \xrightarrow{\text{medie}} X \Rightarrow X_n \xrightarrow{(P)} X$.

Legi ale numerelor mari

T. (T. Bernoulli) legea slabă a nr. mari

A eveniment $P(A)=p$;

$f_n(A)$ frecvența relativă de realizare a lui A în n experiențe independente. atunci avem

$\lim_{n\to\infty} P(|f_n(A) - p| \le \varepsilon) = 1 \quad \forall \varepsilon > 0$

adică $f_n \xrightarrow{(P)} p$ (Kolmogorov = Pascal).

o urnă ⊔ albe. p = prob să extrag una albă
negre.

albă : A ; (putem calcula) 1 albă
0 negră

am avut 10 albe
20 negre.

$p = \frac{10}{30} = \frac{1}{3}$

f_n.

1	$\frac{1}{1}=1$	f_1
0	$\frac{1}{2}$	f_2
0	$\frac{1}{3}$	f_3
1	$\frac{2}{4}$	f_4

(P)

$\frac{1}{3}$

alte teoreme

T-Cebâșev $(X_n)_n$ șir de v.a. independente două câte două cu proprietatea $D^2(X_n) \le c \ \forall n \ge 1$ unde $c \in \mathbb{R}$ fixat

$\Rightarrow (\forall)\ \varepsilon > 0 : \lim_{n\to\infty} P\left(\left|\frac{1}{n}\sum_{j=1}^{n} X_j - \frac{1}{n}\sum_{j=1}^{n} M(X_j)\right| > \varepsilon\right) = 0$

(adică $\frac{1}{n}\sum_{j=1}^{n} X_j \xrightarrow{(P)} \frac{1}{n}\sum_{j=1}^{n} M(X_j)$ $\xrightarrow{(P)} 0$)

$\frac{1}{n}\sum_{j=1}^{n}(X_j - M(X_j)) \xrightarrow{(P)} 0$



T-Poisson A eveniment

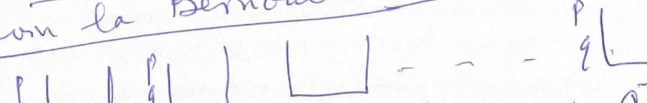
(Schema Poisson)



$\lim_{n\to\infty} P\left(\left|f_n(A) - \frac{p_1 + \dots + p_n}{n}\right| > \varepsilon\right) = 0$.

$f_n(A) - \frac{p_1 + \dots + p_n}{n} \xrightarrow{(P)} 0$

Rezn la Bernoulli - Binomială



n probe, k albe $p_n(k) = \underbrace{C_n^k p^k q^{n-k}}$

$Q(t) = (pt+q)^n = (pt+q)\cdots(pt+q)$

*TLC Teorema limită centrală

• Dacă X v.a. $M(X)$, $D^2(X)$ finite
$\quad\quad\quad

$s_n = \sqrt{\sum_{j=1}^{n} \sigma_j^2}$; $t_n = \sqrt[3]{\sum_{j=1}^{n} \tau_j^3}$

$Y_n = \sum_{j=1}^{n} X_{nj}$, $Z_n = \dfrac{Y_n - M(Y_n)}{s_n}$.

Fie F_n fc. de repartiție ptr Z_n

Z v.a. $\sim N(0,1)$

Dacă $\lim_{n\to\infty} \dfrac{t_n}{s_n} = 0$ atunci

$$\lim_{n\to\infty} F_n(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt.$$

adică: $Z_n \xrightarrow{\text{repr.}} Z$

T. Moivre - Laplace

$(X_n)_n$ iid Bernoulli $X_n : \begin{pmatrix} 1 & 0 \\ p & q \end{pmatrix}$ $q = 1-p$, $p \in (0,1)$

(evident $M(X_n) = p$; $D^2(X_n) = pq$)

$Y_n = X_1 + \dots + X_n$

(aşa că: $M(Y_n) = np$; $D^2(Y_n) = npq$)

Atunci: $\lim_{n\to\infty} P\left(\dfrac{Y_n - np}{\sqrt{npq}} < x\right) = \phi(x)$

unde $\phi(x)$ = fc. de repartiție ptr $N(0,1)$

$$\| \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$$

care fc. de repartiție $\phi(x)$

(sau: $\dfrac{Y_n - np}{\sqrt{npq}} \xrightarrow{\text{repr.}} Z$

* Vectori aleatori

- $X = (X_1, \dots, X_k)$ = vect k-dimensional.
  (v.a, v.a)
  $X : \Omega \to \mathbb{R}^k$
  $(\Omega, \mathcal{K}, P)$ camp prob.
- fc. de repartitie.

$F(x) \in [0,1]$; $x \in \mathbb{R}^k$ $\quad F : \mathbb{R}^k \to [0,1]$.

$F(x) = F(x_1, \dots, x_k) = P(X_1 \le x_1, \dots, X_k \le x_k)$

- $\lim_{x_j \to -\infty} F(x_1, \dots, x_k) = 0 \quad \forall j = \overline{1,k}$
- $\lim_{x_j \to +\infty \ \forall j} F(x_1, \dots, x_k) = 1$
- $\lim_{x_j \to \infty} F_{(x_1, \dots, x_k)}(x_1, \dots, x_{j-1}, x_j, x_{j+1}, \dots, x_k) = F_{(x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k)}(x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k)$

Dacă cap absolut continuu $\Rightarrow \exists$ a densit

$f(x_1, \dots, x_n)$:
$F_X(x_1, \dots, x_k) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_k} f(u_1, \dots, u_k)\, du_1 \dots du_k$

Prop
- $\int_{\mathbb{R}^k} f(x_1, \dots, x_k)\, dx_1 \dots dx_k = 1$
- $f(x_1, \dots, x_k) = \dfrac{\partial^k F(x_1, \dots, x_k)}{\partial x_1 \cdot \dots \cdot \partial x_k}$.

Cap discret apar frame

Caz bidimensional

$X = (X_1, X_2)$; $F_{(X_1,X_2)}$; F_{X_1} , F_{X_2}

caz absolut continuu $\quad f_{(X_1,X_2)} \quad f_{X_1} \quad f_{X_2}$.

$$f_{X_2}(y) = \int_{\mathbb{R}} f_{(X_1,X_2)}(u,v)\,du$$

densitatea lui X_2.

$$\int_{-\infty}^{\infty} f_{(X_1,X_2)}(u,v)\,du$$

marginală

$$f_{X_1}(u) = \int_{-\infty}^{\infty} f_{(X_1,X_2)}(u,v)\,dv$$

densitatea marginală a lui X_1

$$F_{X_1|X_2}(x_1 \mid X_2 = x_2) = \frac{\int_{-\infty}^{x_1} f_{(X_1,X_2)}(u,x_2)\,du}{f_{X_2}(x_2)}$$

$$f_{X_1|X_2=x_2}(x_1 \mid X_2 = x_2) = \frac{f_{(X_1,X_2)}(x_1,x_2)}{f_{X_2}(x_2)}$$

Ex $\quad X_1 \sim \Gamma(a,b)$

$X_2 \mid (X_1 = \alpha) \sim Exp(\alpha)$ } Se cere $f_{X_2}(x_2)$

ols că $f_{(X_1,X_2)}(x_1,x_2) = \underbrace{f_{X_1}(x_1)}_{\text{fgama}} \cdot \underbrace{f(x_2 \mid X_1 = x_1)}_{exp} =$

$$= \frac{a^b}{\Gamma(b)}\,\alpha^{b-1} e^{-\alpha a} \cdot \alpha e^{-\alpha x_2} = \cdots$$

$$f_{X_2}(x_2) = \int_0^{\infty} f_{(X_1,X_2)}(x_1,x_2)\,dx_1 = \cdots$$

Valoare medie pt vector $X = (X_1, \dots, X_k)$

$$M(X_1 X_2 \dots X_k) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_1 \dots x_k f(x_1, \dots x_k)\, dx_1 \dots dx_k$$

$$\left(= \sum_{x_1} \dots \sum_{x_k} x_1 \dots x_k p_{x_1 \dots x_k} \right)$$

- $X = (X_1, \dots, X_k)$
matricea de covariantă $\operatorname{cov}(X_1, \dots, X_k) =$
$= (\operatorname{cov}(X_i, X_j))_{i,j}$ matrice !

- Pt simplitate : X, Y v.a.

$$\operatorname{cov}(X,Y) = M\big((X - M(X))(Y - M(Y))\big)$$

$X = Y \Rightarrow \operatorname{cov}(X,X) = M(X - M(X))^2 = D^2(X)$

$$= M(X^2) - M^2(X)$$

$$\operatorname{cov}(X,Y) = M(XY - X \cdot \underset{ct}{M(Y)} - \underset{ct}{M(X)} \cdot Y + M(X) \cdot \underset{ct}{M(Y)}) =$$

$$= M(XY) - M(Y) \cdot M(X) - M(X) \cdot M(Y) + M(X) \cdot M(Y)$$

$$= M(XY) - M(X) \cdot M(Y)$$

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coeficient de corelație :

$$\rho(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X)} \cdot \sqrt{D^2(Y)}}$$

Avem $|\rho(X,Y)| \leq 1$.

④ Problema 4.
Un împrumut obligatiuni este de 40000 de obligatiuni, fiecare cu valoarea nominală de 1000 u.m. are procentul

Def X, Y sunt necorelate dacă $\rho(X,Y)=0$

Prop Dacă X, Y v. independente $\Rightarrow \rho(X,Y)=0$.

(dem $\rho(X,Y) = \frac{cov(X,Y)}{\sqrt{D^2(X)D^2(Y)}}$ dar

$cov(X,Y) = M(XY) - M(X)M(Y) = 0$

$M(XY) \overset{\text{indep } (X,Y)}{=} M(X)\cdot M(Y)$

Revenim la $X = (X_1, \dots, X_k)$

$cov(X_i, X_j) = M\left((X_i - M(X_i))(X_j - M(X_j))\right)$

$= M(X_iX_j) - M(X_i)\cdot M(X_j)$.

matricea de covariantă:

$$\begin{pmatrix} cov(X_1,X_1) & cov(X_1,X_2) & \cdots & cov(X_1,X_k) \\ cov(X_2,X_1) & cov(X_2,X_2) & \cdots & cov(X_2,X_k) \\ \vdots & & & \\ cov(X_k,X_1) & \cdots & & cov(X_k,X_k) \end{pmatrix}$$

$cov(X_1,X_1)$ dispers.; $cov(X_2,X_2)$ disp.; $cov(X_k,X_k)$ disp.

$cov(X,Y) = cov(Y,X) = M(XY) - M(X)M(Y)$

simetrică

• Medie conditionată

$$M(X \mid Y=y) = \int_{-\infty}^{\infty} x \, f_{X|Y=y}(x|y)\,dx$$

$$M(Y \mid X=x) = \int_{-\infty}^{\infty} y \, f_{Y|X=x}(y|x)\,dy$$

Repartiția normală bidimensională

(X,Y) vector aleator, X, Y v.a. reale

$$f_{(X,Y)}(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left(\frac{(x-m_1)^2}{\sigma_1^2} + \frac{(y-m_2)^2}{\sigma_2^2} - 2\rho\frac{(x-m_1)(y-m_2)}{\sigma_1\sigma_2}\right)\right\}$$

Prop rep. marginale pt X respectiv Y sunt

$$X \sim N(m_1, \sigma_1^2) \; ; \; Y \sim N(m_2, \sigma_2^2)$$

$$\operatorname{cov}(X,Y) = \rho\sigma_1\sigma_2$$

⇒ matricea de covariantă este $\begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$

Prop $(X,Y) \sim$ Bidimensional normal Atunci

$$f_{X|Y}(x|y) = \frac{1}{\sigma_1\sqrt{1-\rho^2}\sqrt{2\pi}} \times$$

$$\times \exp\left\{-\frac{1}{2\sigma_1^2(1-\rho^2)}\left[x-\left(m_1+\rho\frac{\sigma_1}{\sigma_2}(y-m_2)\right)\right]^2\right\}$$

$$f_{Y|X}(y|x) = \frac{1}{\sigma_2\sqrt{1-\rho^2}\sqrt{2\pi}}\left\{\exp\left\{-\frac{1}{2\sigma_2^2(1-\rho^2)}\left[y-\left(m_2+\rho\frac{\sigma_2}{\sigma_1}(y-m_2)\right)\right]^2\right.\right.$$

$\Rightarrow M(X/y) = m_1 + \rho \frac{\sigma_1}{\sigma_2}(y - m_2)$

$M(Y/x) = m_2 + \rho \frac{\sigma_2}{\sigma_1}(x - m_1)$

$D^2(X/y) = \sigma_1^2(1-\rho^2)$, $D^2(Y/x) = \sigma_2^2(1-\rho^2)$

Definiție $x = m_1 + \rho \frac{\sigma_1}{\sigma_2}(y - m_2)$ dreapta de regresie a lui X asupra lui Y.

(dreapta: $y = m_2 + \rho \frac{\sigma_2}{\sigma_1}(x - m_1)$ este dreapta de regresie a lui Y asupra lui X

Prop ! dacă $\rho = 0 \Rightarrow X, Y$ independente

$\Leftarrow$ în general e adevărat

Recapitulare

. Eveniment sigur: Ω ; (Ω, K, P)

Ban: $\Omega = \{S, V\}$

Zar $\Omega = \{1, \dots, 6\}$.

. eveniment $A \subset \Omega$

ban $A = \{S\}$, $A = \Omega$

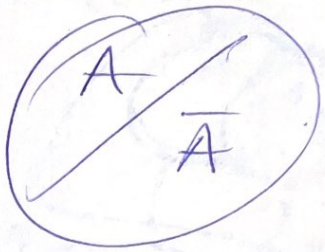
zar $A = \{2, 3\}$, $A = \{1, 6\}$, etc.

. eveniment elementar:

$\{S\}$, $\{V\}$, // $\{i\}$, $i = \overline{1,6}$

• evenimente complementare
A, CA ; A, $\overline{A}$ compl

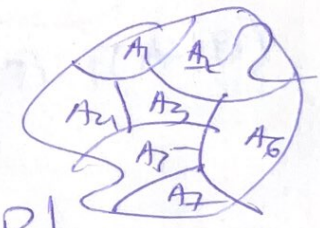
A, B complem
$A \cap B = \emptyset$
$A \cup B = \Omega$



• evenimente (disjuncte): A, B: $A \cap B = \emptyset$.
(incompatibile)

$A_1, \dots, A_n$ incompatibile 2 x 2
$A_i \cap A_j = \emptyset \quad i \neq j$

• partitie a lui Ω
sistem complet de evenimente
$A_1, \dots, A_n \subset \Omega$ ($A_1, \dots, A_n \in \mathcal{K}$)
$A_1 \cup \dots \cup A_n = \Omega$
$A_i \cap A_j = \emptyset \quad \forall i \neq j$



• axiomele prob ($\Omega, \mathcal{K}, P$).
$P: \mathcal{K} \to [0,1]$
• $0 \leq P(A) \leq 1$
• $P(\Omega) = 1$
• $A_1, \dots, A_n$ incompatib 2x2 $\Rightarrow P(A_1 \cup \dots \cup A_n) = P(A_1) + \dots + P(A_n)$.

(în particular A, B incomp $\Rightarrow$
$P(A \cup B) = P(A) + P(B)$.

Evident $P(CA) = P(A^c) = 1 - P(A)$

- formula Poincaré

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

- Formula clasică a prob – B. Pascal

$$p = \frac{\text{nr. caz favorabile}}{\text{nr. caz posibile}}$$

- Probab condiționată

$P(B) \neq 0$; $P(A|B) = \frac{P(A \cap B)}{P(B)}$; $P_B(A)$

$P(A \cap B) = P(B) \cdot P(A|B)$

$P(A) \neq \emptyset$ $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$P(A \cap B) = P(A) \cdot P(B|A)$.

- formula intersecției

$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) =$

$= P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap \dots \cap A_{n-1})$

- A, B sunt independente dacă $P(A \cap B) = P(A) \cdot P(B)$.

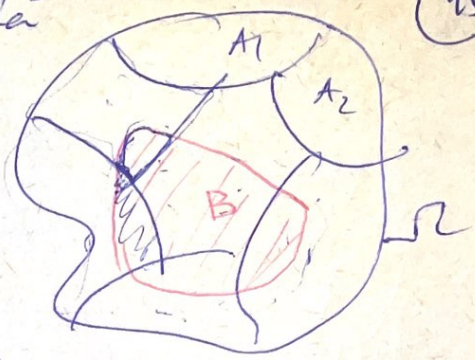
- FPT – formula probabilității totale.

$A_1, A_2, \dots, A_n$ sistem complet de ev/

$\forall B \in \mathcal{K} \Rightarrow P(B) = \sum_{i=1}^{n} P(A_i) \cdot P(B|A_i)$

La rmie la independență

$$P(A|B) = \frac{P(A \cap B)}{P(B)} =$$



$\overset{\text{indep}}{=} \frac{P(A) \cdot P(B)}{P(B)}$

$\Rightarrow \boxed{P(A|B) = P(A)}$

- formula Bayes

$A_1, \dots, A_n$ un sistem complet de evenimente

$B \in \mathcal{K}$, $P(B) \neq 0$ Atunci

$$P(A_j|B) = \frac{P(A_j) \cdot P(B|A_j)}{\sum_{i=1}^{n} P(A_i) \cdot P(B|A_i)}$$

$\parallel P(B)$

(Se folosește $P(A \cap B) = \boxed{P(A) \cdot P(B|A) = P(B) \cdot P(A|B)}$)

★ Variabile aleatoare

v.a discretă: $X = \begin{pmatrix} x_1 & x_2 & \dots & x_i & \dots \\ p_1 & p_2 & \dots & p_i & \dots \end{pmatrix}$

$\begin{cases} p_i \geq 0 \ \forall i \\ \sum_{i \geq 1} p_i = 1 \end{cases}$ condiție de normalizare

repartiția v.a. X

$p_i = P(X = x_i) = P(\{\omega \in \Omega \mid X(\omega) = x_i\}) \in \mathcal{K}$

- fc. de repartiție

Cu grijă Vezi Curs 2

$F(x) = P(X < x)$

" sau " $P(X \leq x) = \sum_{i/x_i \leq x} p_i$! de revedeti

$M(X) = \sum_{i \geq 1} x_i p_i = m_x$

dacă $g \to g(X)$; $M(g(X)) = \sum g(x_i) \cdot p_i$

$i \geq 1$

• $g(X) : \begin{pmatrix} g(x_1) & g(x_2) \dots g(x_i) \dots \\ p_1 & p_2 \dots p_i \dots \end{pmatrix}$

• Dispersia (sau variația)

$\sigma_X^2 = \sigma^2(X) = D^2(X) = Var(X) = M(X^2) - (M(X))^2$

$X^2 : \begin{pmatrix} x_1^2 \dots x_i^2 \dots \\ p_1 \dots p_i \dots \end{pmatrix}$

• abaterea medie pătratică $\sigma = \sqrt{D^2(X)}$

• momentul de ordin r în raport cu media

$M((X - m_x)^r) = \sum_{i \geq 1} (x_i - m_x)^r \cdot p_i$

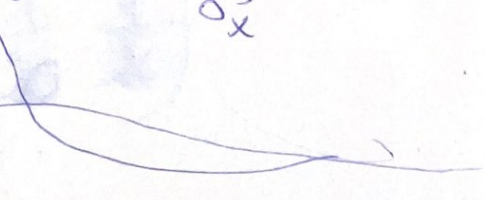
• moment de ordin r:

$M(X^r) = \sum_{i \geq 1} x_i^r \cdot p_i$

• coeficient de asimetrie: $\gamma_1 = \frac{M(X - m_x)^3}{\sigma_X^3}$

• coeficient de aplatizare

$\gamma_2 = \frac{M(X - m_x)^4}{\sigma_X^4} - 3$



• Cuantile (Q_α) $\alpha \in [0,1]$.
cuantilă de ordin α pt X dacă

$$P(X < Q_\alpha) \le 1-\alpha \le P(X \le Q_\alpha)$$

(!) la caz discret!

Caz particular: mediana: $Q_{1/2} = Q_{0.5} = me$

$$P(X < me) \le \frac{1}{2} \le P(X \le me)$$

(n-a. X: 2 5 1 3 4 1 5 2 3
x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9

ordonăm crescător
1 1 2 2 [3] 3 4 5 5
$x_{(1)}$ $x_{(2)}$ $x_{(3)}$ $x_{(5)}$

statistica de ordine

dacă 1 1 2 2 [3 4] 4 4 5 5
$x_{(1)}$ $x_{(10)}$

$me = \frac{3+4}{2} = 3.5$)

Ex. Valoare modală: x_0 : pt care cea mai mare valoare.

$$X : \begin{pmatrix} 2 & 5 & 1 \\ \frac{1}{9} & \frac{2}{9} & \frac{6}{9} \end{pmatrix}$$

$x_0 = 1$
val modală

prob max!

$$X : \begin{pmatrix} 2 & 5 & 1 & 7 & 3 \\ \frac{3}{10} & \frac{3}{10} & \frac{3}{10} & \frac{1}{20} & \frac{1}{20} \end{pmatrix}$$

prob max.

x_0 val modă $\in \{1,2,5\}$.

· fc. generatoare.

$$G_X(t) = M(e^{tX}) = \sum_{j\geq 1} e^{tx_j} \cdot p_j$$

· fc. caracteristică

$$\varphi_X(t) = M(e^{itX}) = \sum_{j\geq 1} e^{itx_j} \cdot p_j$$

$i = \sqrt{-1}$

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* Caz v.a. continue

aici $P(X<x) = P(X\leq x)$

(pt că $P(X=x)=0$)

abs. continuă : f densitate prob.

$f \geq 0$

$\int_{-\infty}^{\infty} f(x)\,dx = 1$ (cond. de normalizare)

fc. repartiție $F(x) = \int_{-\infty}^{x} f(t)\,dt$

$F'(x) = f(x)$.

· val medie $M(X) = \int_{-\infty}^{\infty} x f(x)\,dx = m_X$

· disp. $D^2(X) = \int_{-\infty}^{\infty} (x-m_X)^2 f(x)\,dx$

$= M(X^2) - (M(X))^2$

$M(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$

· Moment ord r $\nu_r = M(X^r) = \int_{-\infty}^{\infty} x^r f(x)\,dx$

· γ_1, γ_2 se definesc la fel.

· <u>cuantila</u>: deoarece $P(X=x)=0$

avem: Q_α cuantilă ordin α dacă

$$P(X \le Q_\alpha) = 1-\alpha.$$

· mediană: $\alpha = 1/2$ // $F(m_e) = \frac{1}{2}$

$$F(Q_\alpha) = 1-\alpha \quad \| \quad \alpha = 1/2$$

· val. modală val x_0 pt care f e max

· fc. generatoare $G_X(t) = M(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$

· fc. caract: $\varphi_X(t) = M(e^{itX}) = \int_{-\infty}^{\infty} e^{itx} f(x)\,dx$

★ Vector aleator

· <u>Caz discret</u> X valori $x_1 \dots x_i \dots$

Y $\dots y_1 \dots y_j \dots$

$(X,Y) = ((x_i, y_j))_{i,j}$

(X,Y) <u>repartiție</u> $\begin{pmatrix} (x_i,y_j) \\ p_{ij} \end{pmatrix}_{i,j}$

$$\begin{pmatrix} (x_1,y_1) \dots (x_1,y_j) \dots (x_2,y_1) \dots (x_2,y_j) \dots \\ p_{11} \quad p_{1j} \dots \quad p_{ij} \end{pmatrix}$$

$p_{ij} = P(X = x_i, Y = y_j) =$

$= P\left(\{\omega \in \Omega / X(\omega) = x_i\} \cap \{\omega \in \Omega / Y(\omega) = y_j\}\right)$

$A_i \cap B_j$ — X, Y indep. daca

$= P(A_i) \cdot P(B_j) =$

$= P(X = x_i) \cdot P(Y = y_j) =$

$= p_{i\cdot} \times p_{\cdot j}$ aici pp

$X \backslash Y$	y_1	$\cdots$	y_j	$\cdots$	y_m	
x_1	p_{11}		p_{1j}		p_{1m}	$p_{1\cdot}$
$\vdots$						$\vdots$
x_i	p_{i1}		p_{ij}		p_{im}	$p_{i\cdot}$
$\vdots$						$\vdots$
x_n	p_{n1}		p_{nj}		p_{nm}	$p_{n\cdot}$
	$p_{\cdot 1}$		$p_{\cdot j}$		$p_{\cdot m}$	

$p_{i\cdot} = \sum_{j=1}^{m} p_{ij}$ $\quad$ $p_{\cdot j} = \sum_{i=1}^{n} p_{ij}$

$Y : \begin{pmatrix} y_1 & \cdots & y_j & \cdots & y_m \\ p_{\cdot 1} & & p_{\cdot j} & & p_{\cdot m} \end{pmatrix}$ repartitia marginala a lui Y

$X : \begin{pmatrix} x_1 & \cdots & x_i & \cdots & x_n \\ p_{1\cdot} & & p_{i\cdot} & & p_{n\cdot} \end{pmatrix}$ rep. marginala a lui X

$M(X) = \sum_{i=1}^{m} x_i p_{i\cdot}$, $M(Y) = \sum_{j=1}^{n} y_j p_{\cdot j}$

$M(XY) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j p_{ij}$

$P(X = x_i / Y = y_j) = \frac{P(X = x_i, Y = y_j)}{P(Y = y_j)} = \frac{p_{ij}}{p_{\cdot j}}$

$p_{i/j} = \frac{p_{ij}}{p_{\cdot j}}$ $\qquad$ $p_{j/i} = \frac{p_{ij}}{p_{i\cdot}}$

$X / Y = y_j : \begin{pmatrix} x_1 & \cdots & x_i & \cdots & x_m \\ p_{1/j} & \cdots & p_{i/j} & \cdots & p_{m/j} \end{pmatrix}$

la fel $Y / X = x_i \begin{pmatrix} y_1 & \cdots & y_j & \cdots & y_n \\ \cdots & \cdots & p_{j/i} & \cdots & \cdots \end{pmatrix}$

rep. lui Y / conditionată de $X = x_i$

Pot calcula $M(X / Y = y_j) = \sum_{i=1}^{m} x_i \cdot p_{i/j}$

$M(Y / X = x_i) = \sum_{j=1}^{n} y_j \cdot p_{j/i}$

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* Caz continuu

caz absolut continuu $\Rightarrow \exists$ densitate $f(x,y)$ a vectorului (X,Y)

$f_{(X,Y)} \geq 0$, $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dx\,dy = 1$

fc. repartitie

$F_{(X,Y)}(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{(X,Y)}(x,y)\,dx\,dy$

densitatea marginala pt X

$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

dens. marginala pt Y: $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$.

$F_X(x) = \lim_{y \to \infty} F_{(X,Y)}(x,y)$

- X, Y in dependente daca $f_{(X,Y)}(x,y) = f_X(x) \cdot f_Y(y)$ $\forall x, y$.

sau

$F_{(X,Y)}(x,y) = F_X(x) \cdot F_Y(y)$

- densitate conditionata

$f_{X|Y=y}(x/y) = \frac{f(x,y)}{f_Y(y)}$ // $f_{Y|X=x}(y/x) = \frac{f(x,y)}{f_X(x)}$

Pot calcula

$M(X|Y=y) = \int_{-\infty}^{\infty} x \cdot f_{(X|Y=y)}(x/y)\,dx$.

$M(Y|X=x) = \int_{-\infty}^{\infty} y \cdot f_{(Y|X=x)}(y/x)\,dy$.

Ex (X,Y) vect.

$$f_{(X,Y)}(x,y) = \begin{cases} c(x+y+1), & x \in [0,1],\ y \in [0,2] \\ 0 & \text{în rest.} \end{cases}$$

a) $c=?$ f densit.
b) densit marginale
c) X, Y indep?
d) fc. rep. marginale; fc. rep (X,Y)
e) $f_{X|Y=y}$, $f_{Y|X=x}$, $M(X|Y=y)$, $M(Y|X=x)$

dem $f \geq 0 \Rightarrow \boxed{c>0}$ pt că $x+y+1 \neq 0$ $\forall x,y \in$ [...]

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\,dx\,dy = 1 \qquad c\int_0^1\int_0^2 (x+y+1)\,dx\,dy =$$

$$= c\left[\int_0^1\left[\int_0^2 x\,dx\right]dy + \int_0^1\left(\int_0^2 y\,dx\right)dy + \int_0^1\int_0^2 dx\,dy\right]$$

$$= c\left[\left(\int_0^1 x\,dx\right)\cdot\left(\int_0^2 dy\right) + \int_0^1 dx\int_0^2 y\,dx + \int_0^1 dx\int_0^2 dy\right] = c\left[\frac{x^2}{2}\Big|_0^1 \cdot y\Big|_0^2 + x\Big|_0^1 \cdot \frac{y^2}{2}\Big|_0^2 + x\Big|_0^1 \cdot y\Big|_0^2\right]$$

$$= c\left[\frac{1}{2}\cdot 2 + 1\cdot 2 + 1\cdot 2\right] = 5c = 1 \Rightarrow \boxed{c = \frac{1}{5}}$$

$f_{(X,Y)}(x,y) = \begin{cases} \frac{1}{5}(x+y+1), & x \in [0,1],\ y \in [0,2] \\ 0 \text{ rest} \end{cases}$ (24)

$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dy = \frac{1}{5}\int_0^2 (x+y+1)\,dy =$

$= \frac{1}{5}\left[x\cdot\int_0^2 dy + \int_0^2 y\,dy + \int_0^2 dy\right] =$

$= \frac{1}{5}\left[x\cdot y\Big|_0^2 + \frac{y^2}{2}\Big|_0^2 + y\Big|_0^2\right] =$

$= \frac{1}{5}\left[2x + 2 + 2\right] = \frac{2}{5}(x+2)$

$x \in [0,1]$

$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dx =$

$= \frac{1}{5}\int_0^1 (x+y+1)\,dx = \frac{1}{5}\left[\int_0^1 x\,dx + y\int_0^1 dx + \int_0^1 dx\right]$

$= \frac{1}{5}\left[\frac{x^2}{2}\Big|_0^1 + y\cdot x\Big|_0^1 + x\Big|_0^1\right] =$

$= \frac{1}{5}\left[\frac{1}{2} + y + 1\right] = \frac{1}{5}\left(y + \frac{3}{2}\right)$

$y \in [0,2]$

X, Y indep dacă:

$f_{(X,Y)}(x,y) \overset{?}{=} f_X(x)\cdot f_Y(y)$

$\frac{1}{5}(x+y+1) \overset{?}{=} \frac{2}{5}(x+2)\cdot\frac{1}{5}\left(y+\frac{3}{2}\right)$

nu sunt indep.